

Saurabh Jalendra

saaurabhjalendra@gmail.com

+91-9783982649

Jaipur, India (open to relocate)

 LinkedIn

 GitHub

Research Interests

Reinforcement learning, world models, scientific simulation and discovery, symbolic regression, multimodal learning, LLM agents

Education

Birla Institute of Technology and Science, Pilani – M.Tech in AI/ML

Jun 2024 – Mar 2026

- **Dissertation:** Quantum-Enhanced Simulation Learning for RL
- **Coursework:** Deep Reinforcement Learning, Computer Vision, NLP, Probabilistic Graphical Models, Graph Neural Networks

NMIMS, Mumbai – MBA in Banking and Finance

Jan 2022 – Jan 2024

Government Engineering College, Bikaner – B.Tech in Computer Science Engineering

May 2013 – May 2017

Experience

Co-Founder & AI Lead – SKY AI, Jaipur

Oct 2024 – Present

- Built **Klares**: Document intelligence platform for Hong Kong finance with DDQ autofill (LlamaIndex RAG for filling unfilled DDQs using scoped documents), SBERT document categorization, vector-based knowledge base (pgvector + cross-encoder re-ranking), and AI-extracted compliance reminders with email notifications.
- Led development of **International Citizen**: AI-powered portfolio management platform with automated statement extraction, multi-currency aggregation, and conversational AI assistant for global investors.

Research Intern – ISRO (National Remote Sensing Centre), Jodhpur

Feb 2017 – Jun 2017

- Developed VR heritage visualization application using Unity deployed on Oculus Rift and HTC Vive platforms.
- Built photogrammetry pipeline with Meshroom generating high-fidelity 3D models from aerial drone imagery.

Selected Projects

Simulating Anything 

Feb 2026 – Present

- Domain-agnostic scientific discovery pipeline: natural language query → simulation → RSSM world model (JAX/Equinox) → symbolic regression → human-interpretable equations.
- Implemented 187 simulation domains across physics, biology, and chemistry; PySR/SINDy recovered known physical laws with $R^2 \geq 0.999$ on 11/14 core domains (mean $R^2 = 0.970$); benchmarked on NVIDIA RTX 5090.
- 570 cross-domain mathematical isomorphisms detected across 125 mathematical equivalence classes.

Quantum-Enhanced Simulation Learning for RL 

Nov 2025 – Feb 2026

- M.Tech dissertation (defended) comparing quantum-inspired vs classical training for DreamerV3-style world models.
- Implemented 4 quantum-inspired methods (tunneling optimizer, superposition buffer, entanglement layer, interference ensemble) benchmarked against classical baseline across 8 environments: CartPole, Pendulum, Walker-walk, Cheetah-run, Reacher-easy/hard, Pong, Breakout.
- Interference Ensemble achieved 36–47% reduction in world model prediction MSE on DMControl robotics (Walker, Cheetah, Reacher-easy/hard; $p < 0.008$, Mann-Whitney U with Bonferroni correction); 200 experiments across 5 methods × 5 seeds × 8 environments.

Multimodal Gravitational Lensing Classification 

Mar 2025

- Built multimodal architecture based on Neural Computing and Applications (2023): 4-block CNN (4-channel, 45×45 images) + 2-layer LSTM (14-step light curves) with late-fusion classification head; 722K trainable parameters.
- Trained on 20K DES-deep survey samples across 4 balanced gravitational lensing classes; 76% F1 on non-lens detection.

Analyst Coder-Agent 

Nov 2025

- Autonomous LLM agent with tool calling (file ops, code search, git, reporting) via FastAPI + React + Claude 3.5 Sonnet; plan-approval safety workflow with JWT auth, PostgreSQL session persistence, and Docker Compose deployment (4 services).

CT-MRI Fusion with Cross-Modal Attention 

Oct 2025

- Dual-branch ResNet50 fusion model with cross-modal attention gates for integrating CT and MRI imaging data; trained on 4,974 images with mixed-precision optimization (FP16 + TF32) on NVIDIA RTX 4050.

GNN Link Prediction on OGB-Collab 

Sep 2025

- Benchmarked GCN, GraphSAGE, and GAT on the ogbl-collab dataset (235K nodes, 2.3M edges); GCN achieved best Hits@50 of 5.59% using official OGB evaluation with 100K negative samples per edge.

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, SQL
- **ML/DL:** PyTorch, JAX/Equinox/Optax, TensorFlow/Keras, scikit-learn, XGBoost, PyTorch Geometric, PySR/PySINDy, Gymnasium, DMControl, OpenCV
- **Data/OCR:** Pandas, NumPy, Matplotlib, Seaborn, pdfplumber, Docling, RapidOCR
- **LLM/NLP:** LangChain, LlamaIndex, Hugging Face, SBERT, Claude API, Gemini API, OpenRouter, RAG pipelines
- **Web/Backend:** FastAPI, Node.js, Next.js, React, Streamlit, PostgreSQL, Redis, SQLite
- **Cloud/DevOps:** AWS (S3, EC2, RDS), GCP, Docker, GitHub Actions, CI/CD